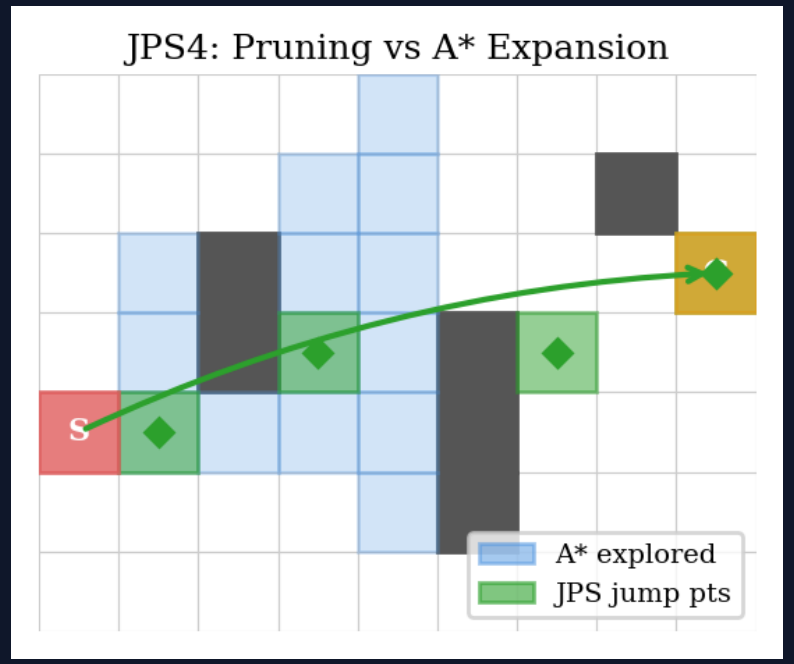
HABIB UNIVERSITY | ALGORITHM DESIGN & ANALYSIS

Comparing Shortest-Path Algorithms on Grid Maps

Dijkstra vs A* vs JPS4 on 4-connected grids

Dijkstra **A*** **JPS4**



Interim Progress Report

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Three Algorithms, One Problem

Shortest path on a 4-connected grid. Same input, very different behavior.

DIJKSTRA

The baseline

Explores every reachable cell in order of distance from start. Always correct, but has no sense of where the goal is, so it essentially maps the entire reachable area first.

Mechanism

Uniform wavefront

A*

Heuristic guided

Adds Manhattan distance to the goal as a lower-bound estimate. Picks the cell with the lowest total estimated cost at each step, so the search moves toward the goal instead of spreading evenly.

Mechanism

Goal-biased search

JPS4

Baum (2025)

Skips over long open corridors entirely. Stops only at jump points, places where an obstacle forces a real routing choice. Only those cells enter the open list, so the list stays small.

Mechanism

Symmetry pruning

All three guarantee the same shortest path. The difference is how many cells each one has to look at.

Methodology

Same benchmark, same grids, same search loop. Only the algorithm swaps.

Benchmark Setup

GRID

50 x 50 random obstacles

DENSITIES

15 percent, 28 percent, 40 percent

TRIALS PER DENSITY

24 accepted runs

SEED

42 (deterministic)

START AND GOAL

Placed far apart, verified reachable

ACCEPTANCE RULE

Trial counted only if all three algorithms succeed

Shared Search Loop

One A* engine. Heuristic and successor function plug in per algorithm.

astar_jps( )

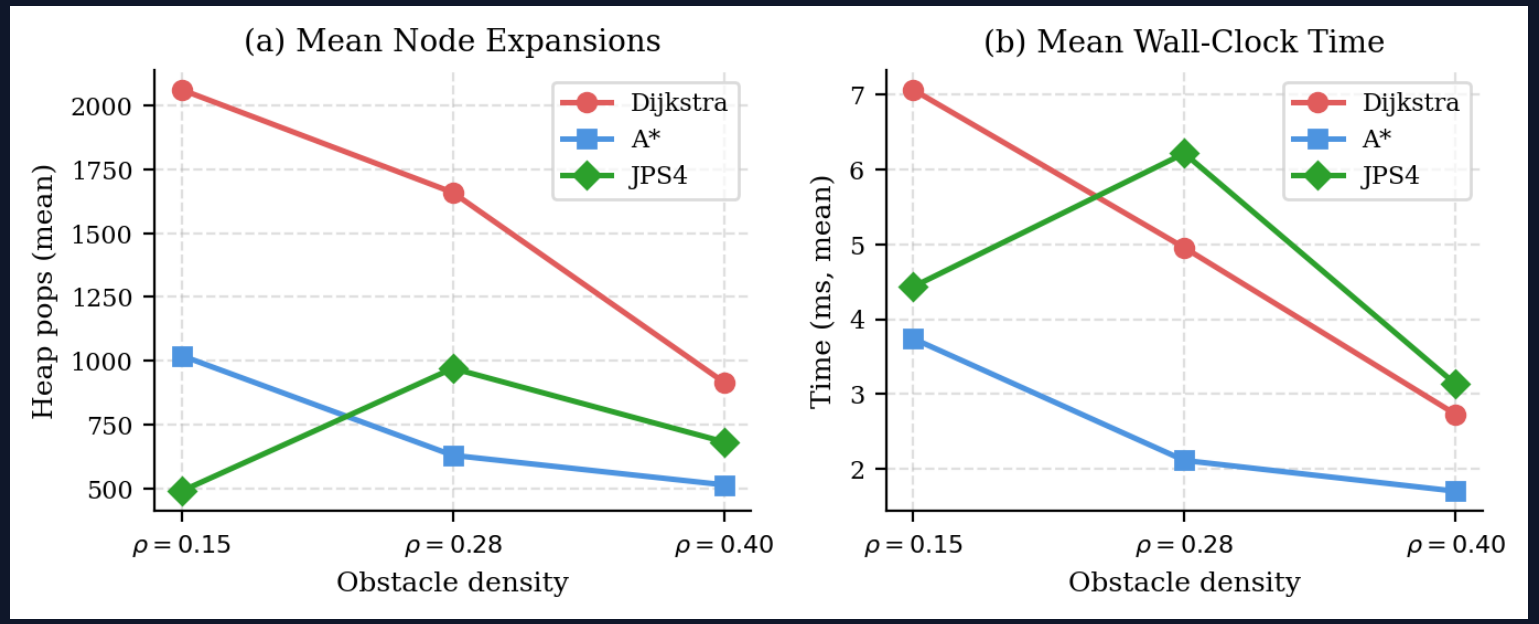
heap-based open list

Dijkstra	A*	JPS4
h = 0	h = Manhattan	h = Manhattan
4-neighbour successors	4-neighbour successors	jump-point successors

Any measured difference comes from the algorithm itself.

Preliminary Results

Node expansions and wall-clock time vs. obstacle density. 24 trials each.



Key Findings

A* halves Dijkstra

Around 50 percent of Dijkstra's expansions at every density. Manhattan heuristic pays off consistently.

JPS4 wins on open maps

At 15 percent density, JPS4 visits only 488 nodes versus 2062 for Dijkstra. A 76 percent reduction.

JPS4 loses on dense maps

At 28 percent, JPS4 expands 970 cells versus A*'s 629. Shorter corridors mean shorter jumps.

CORRECTNESS

All three return equal path lengths on every trial.